Adebanji Oluwatimileyin Adelowo

Machine Learning Engineer | Data Scientist | Scientific Computing

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Profile

Mathematical Engineer and Machine Learning Engineer with experience in scientific computing, inverse problems, numerical PDEs, signal processing, and applied machine learning. Research background in optimization and fluid dynamics combined with industry experience building ML and LLM systems for aviation telemetry, predictive analytics, and computer vision. Interested in scientific machine learning, computational mathematics, and AI-driven engineering systems.

Professional Experience

Data Scientist — *Dec 2024 – Present*
Intuos Srl — *L'Aquila, Italy*

- Developed ML Classifiers (Random Forest and LGBMClassifier) for real-time flight phase identification
- Developed FFT-based signal processing pipelines for aircraft engine RPM estimation and anomaly detection
- Built REST APIs and automated telemetry data processing pipelines
- Developed dashboards for operational monitoring and flight performance analysis

Machine Learning Engineer — *Apr 2023 – Nov 2024*
N-and Italia Srl — *Pognano, Italy*

- Developed predictive ML models for vending machine delivery failure forecasting
- Processed and analyzed large-scale operational datasets containing transaction and telemetry logs
- Built dashboards tracking KPIs, user behavior metrics, and machine reliability statistics
- Performed statistical analysis of system performance and operational anomalies

Operational Data Analyst — *Jan 2023 – Mar 2023*
eResult Srl — *Cesena, Italy*

- Optimized SQL queries and reporting workflows for business intelligence systems
- Improved database performance and automated reporting pipelines

Digital Engineering Intern — *Apr 2022 – Oct 2022*
Bridgestone EMIA — *Rome, Italy*

- Improved efficiency of `deepcopy` on `GeometryMap` half-edge data structures used in tire cross-section FEA meshing pipelines
- Analysed a multi-stage tire assembly build process (`GenericTAM`: BeadAssembly, FirstStage, SecondStage, Tread, TurnUp) to identify copy bottlenecks across nested Bézier-backed geometry objects
- Developed geometric algorithms for cubic Bézier curve closest-point computation using scalar minimization

Machine Learning Intern — *May 2022 – Aug 2022*
Fellowship.AI — *San Francisco, CA (Remote)*

- Engaged in deep learning research focusing on convolutional networks and Generative Adversarial Networks (GANs)
- Implemented InsetGAN architecture for whole-body human image generation and deployed the inference service to a Swift iOS app via FastAPI

Technical Skills

Programming
Python, SQL, MATLAB, C++, R, LaTeX

Machine Learning & AI
PyTorch, TensorFlow, scikit-learn, LangChain, OpenAI

Data Engineering & APIs
FastAPI, Docker, PostgreSQL, SQL Server, Git

Signal Processing
FFT analysis, anomaly detection

Numerical Methods
Finite differences, finite elements, finite volumes

Visualisation
Matplotlib, Seaborn, Plotly, Power BI

Cloud & DevOps
AWS, Google Cloud

Selected Projects

INTUOS Flight Data Monitoring Dashboard — Full-stack aviation safety analytics platform; FastAPI + IBM DB2 backend with React/Vite frontend. Real-time telemetry visualisation, flight phase classification, and engine performance metrics. *Stack: FastAPI, IBM DB2, React, Vite, Docker*

Aircraft Engine RPM Estimation via Audio DSP — Real-time engine health monitoring on Raspberry Pi; FFT-based RPM extraction from microphone audio transmitted over UART to ESP32. Validated against Tecnam P92 / Rotax 912.

Vending Machine Failure Prediction — Predictive ML models for delivery failure forecasting across 150+ machines; pre-failure event sequence detection from transaction and telemetry logs.

LLM Engineering Portfolio — End-to-end AI systems: Medical RAG Assistant, Enterprise NL-to-SQL engine, and Financial Sentiment Distillation system. *Stack: LangChain, OpenAI, Pinecone, FastAPI*

Image Denoising — ResNet with Residual Channel Attention Blocks (RCAB) and Squeeze-and-Excitation for grayscale image denoising; blind denoising with random σ, SSIM+L1 loss, achieving $\sim$26–28 dB PSNR on LFW.

Multi-Label Image Classification — Multi-label classification on Pascal VOC 2012 (20 classes) using EfficientNet-B3 with mixed-precision training, OneCycleLR, and mAP evaluation.

Education

M.Sc. in Mathematical Engineering — *2018 – 2021*
University of L'Aquila — *Italy*
Erasmus Mundus MathMods Scholarship

B.Sc. in Mathematics — *2014 – 2018*
Obafemi Awolowo University — *Nigeria*

Honors and Awards

- **Erasmus Mundus MathMods Scholarship** — European Commission (2018–2021)
- **MTN Foundation, PTDF, and Murli T. Chellaram Scholarships** — Merit-based undergraduate awards (2014–2018)